PHYSICAL SCIENCES

PHYSICAL SCIENCES GRADE 12 SOURCE-EMBEDDED SMOKE TEST

SUBJECT	PHYSICAL SCIENCES
GRADE	12
ASSESSMENT TASK	PHYSICAL SCIENCES GRADE 12 SOURCE-EMBEDDED SMOKE TEST
MARKS	27
TIME	45 minutes

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.

2. Answer all questions or complete all activities as instructed.

3. Show all working where calculations, planning, diagrams or explanations are required.

4. Use the space provided by the educator, answer booklet or learner task book.

5. Write neatly and clearly.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Data Table

1.7 The table below shows the ionisation constants, K_a, for two acids at 25 °C.

ACID	K_a
Butanoic acid	$1,5 \times 10^{-5}$
Ethanoic acid	$1,8 \times 10^{-5}$

Consider the following statements for these two acids when they have equal concentration at 25 °C:

(i) Both are weak acids.
(ii) Butanoic acid is a stronger acid than ethanoic acid.
(iii) The butanoic acid solution has a lower concentration of hydronium ion, $H_3O^+(aq)$, than the ethanoic acid solution.

Which of the above statements are TRUE?

A (i) and (ii) only

B (i) and (iii) only

C (ii) and (iii) only

D (i), (ii) and (iii) (2)

1.8 Which ONE of the following pairs of acids and bases, all of the same concentration, react to give the highest pH at the equivalence point in a titration at 25 °C?

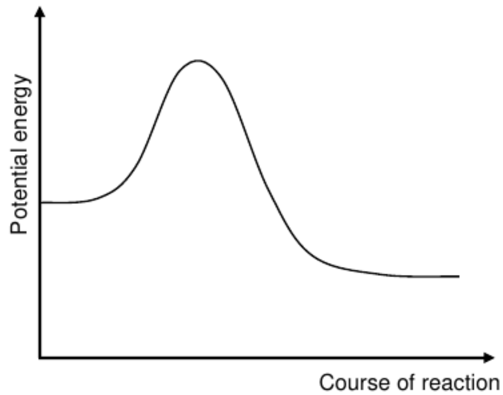
A HCℓ and NH_3

B HCℓ and NaOH

SOURCE 2: Official-paper exemplar: Diagram Or Model

1.4 The potential energy diagram below is for the following hypothetical chemical reaction:

$$X + Y \rightleftharpoons Z$$



Which ONE of the following combinations of values for the heat of the reaction and the activation energies can be obtained for this reaction?

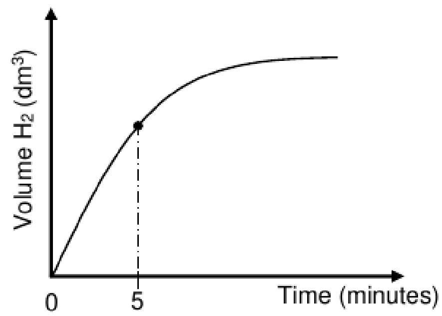
	$\Delta H_{(forward)}$ ($kJ\cdot mol^{-1}$)	**$E_{A(forward)}$ ($kJ\cdot mol^{-1}$)**	**$E_{A(reverse)}$ ($kJ\cdot mol^{-1}$)**
A	-400	300	100
B	-200	300	100
C	+400	100	300
D	-200	100	300

(2)

1.5 Initially, an equal number of moles of hydrogen gas, $H_2(g)$, and iodine gas, $I_2(g)$, are mixed in a closed container. The reaction reaches equilibrium at a

SOURCE 3: Official-paper exemplar: Graph Or Chart

The graph of volume $H_2(g)$ versus time for this experiment, not drawn to scale, is shown below.



5.1.2 For the time interval t = 0 to t = 5 minutes, the average reaction rate for the formation of $H_2(g)$ is 0,033 $dm^3 \cdot min^{-1}$.

Calculate the mass of Aℓ present in the container at t = 5 minutes. Take the molar gas volume as 24,5 $dm^3 \cdot mol^{-1}$. (6)

Assume that the concentration of the HCℓ(aq) stays constant for the duration of the reaction.

5.1.3 Use the collision theory to explain the change in the reaction rate from t = 0 to t = 5 minutes. (4)

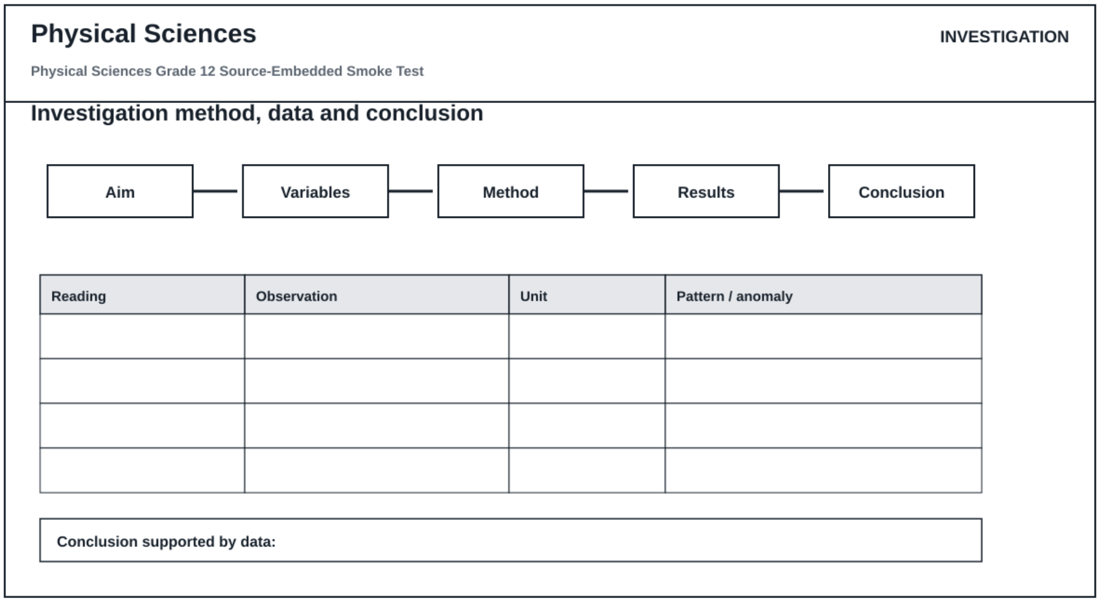
EXPERIMENT II

Experiment I is repeated using a 2 $mol \cdot dm^{-3}$ HCℓ solution.

5.1.4 Redraw the above graph (NO numerical values need to be shown)

VISUAL MATERIAL

FIGURE 1: Investigation Data And Method Sheet



Physical Sciences

INVESTIGATION

Physical Sciences Grade 12 Source-Embedded Smoke Test

Investigation method, data and conclusion

Aim — Variables — Method — Results — Conclusion

Reading	Observation	Unit	Pattern / anomaly

Conclusion supported by data:

QUESTIONS

QUESTION 1: SOURCE INTERPRETATION

Instructions: Use the supplied source material. Answer from visible evidence and relevant subject knowledge.

1.1. Refer to SOURCE 1. Identify one important feature of the data table and explain how it could be used as evidence in Physical Sciences. **(4)**

1.2. Refer to SOURCE 2. Identify one important feature of the diagram or model and explain how it could be used as evidence in Physical Sciences. **(4)**

1.3. Refer to SOURCE 3. Identify one important feature of the graph or chart and explain how it could be used as evidence in Physical Sciences. **(4)**

1.4. Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why. **(5)**

QUESTION 2: METHOD AND REVIEW

Instructions: Evaluate whether this generated assessment pack is ready for a real learner group.

2.1. State TWO checks an educator must perform before releasing this assessment to learners. **(4)**

2.2. Explain why official-paper source exemplars are stronger than generic generated visuals for this route. **(6)**

MARKING GUIDELINE

TASK 1: SOURCE INTERPRETATION

1.1. Refer to SOURCE 1. Identify one important feature of the data table and explain how it could be used as evidence in Physical Sciences. **(4)**

- Relevant feature from SOURCE 1 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.2. Refer to SOURCE 2. Identify one important feature of the diagram or model and explain how it could be used as evidence in Physical Sciences. **(4)**

- Relevant feature from SOURCE 2 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.3. Refer to SOURCE 3. Identify one important feature of the graph or chart and explain how it could be used as evidence in Physical Sciences. **(4)**

- Relevant feature from SOURCE 3 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.4. Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why. **(5)**

- Two supplied sources are compared.
- A justified judgement is made.
- Reasoning mentions readability, data richness, visual clarity or subject fit.

TASK 2: METHOD AND REVIEW

2.1. State TWO checks an educator must perform before releasing this assessment to learners. **(4)**

- Check source readability and relevance.
- Check CAPS topic/grade alignment.
- Check mark allocation and memo correctness.
- Check copyright/reuse conditions for source material.

2.2. Explain why official-paper source exemplars are stronger than generic generated visuals for this route. **(6)**

- They reveal actual source types used in exams.
- They reduce invented visual nonsense.
- They still require adaptation and educator approval.

RUBRIC

Criterion	Marks	Descriptor
Source evidence use	10	Answers use visible evidence from supplied sources.
Subject accuracy	10	Answers use correct subject concepts and terminology.
Review readiness	10	Risks and approval checks are identified clearly.

EDUCATOR REVIEW CHECKLIST

- Confirm each embedded source is readable after PDF export.
- Confirm source use/copyright status before client delivery.
- Confirm questions refer only to supplied or clearly stated evidence.
- Confirm memo and mark allocation match the subject route.